Solution Design Document — Document Management System (DMS)

Oct 21, 2026 · School ERP, Pamantasan ng Lungsod ng Pasig · Version 1.0 (draft)

1. Document control

Version history

Version	Date	Author	Change summary
0.1	2026-10-10	Thomas Dominique Baran	Architecture diagrams drafted (5 pages)
0.2	2026-10-13	Thomas Dominique Baran	Software design document drafted; scope set as the ERP document repository; students out of scope
0.3	2026-10-13	Thomas Dominique Baran	Restructured into the Solution Design Document format; reviewers and executive summary added
0.4	2026-10-13	Thomas Dominique Baran	Team edits: objectives, scope, FR-03, NFR-03; decisions log AD-01 to AD-07 (local ClamAV scanning is AD-01); architecture figures added

0.5	2026-10-18	Thomas Dominique Baran	Aligned with the 18 Oct implementation plan: Super Admin, Admin and Staff personas; invitation-only accounts; approvals; workspace registry; OAS-to-ES packet release; delivery status updated
0.6	2026-10-21	Thomas Dominique Baran	Availability, recovery, audit retention, accessibility and capacity targets filled; persona estimates added
1.0	2026-10-21	Thomas Dominique Baran	Version history updated; proposed owners named for migration tasks
1.1	2026-10-23	Thomas Dominique Baran	Scope reduction: Removed PMS, EMS, and ES document custody. DMS is now strictly the OAS-to-ES bridge and internal workspace. Dropped FR-11 (Memos).

Reviewers and approvers

Name	Role	Organisation	Responsibility
Joseph Wilfred Dela Cruz	Client	Pamantasan ng Lungsod ng Pasig (Client)	Reviewer: Client Requirements

Niccolo Yumul	Enterprise Architect	Malunggay Pandesal (Employee Management System)	Approver: Enterprise Solutions
Prespurt Bellen	Solutions Architect	Malunggay Pandesal (Employee Management System)	Reviewer: EMS Solution
Nathan Fernandez	Solutions Architect	Antonio Pigafetta (Payroll Management System)	Reviewer: PMS Solution
Thomas Baran	Solutions Architect	Nagpayong Pinagbuhatan (Document Management System)	Reviewer: DMS Solution
Neil Aguilar	Solutions Architect	(Enrollment System)	Reviewer: ES Solution
Gabriel Abella	Solutions Architect	Avisala (Online Admission System)	Reviewer: OAS Solution

Related documents: Architecture diagrams (ARCHITECTURE.drawio); Business Requirements Document, DPIA, Test Strategy and Cutover Plan.

2. Executive summary

The school ERP for Pamantasan ng Lungsod ng Pasig is made of four independent systems: Employee Management (EMS), Payroll Management (PMS), Enrollment (ES) and Online Admission (OAS).

There is currently no secure, controlled place to hold cross-system documents. We propose the Document Management System (DMS) as a centralized repository and secure bridge. It is one Laravel 12 application with its own MySQL database and private file storage.

The primary cross-system use case is the secure transfer of verified applicant documents from OAS to ES, eliminating the need for students to resubmit requirements. The four systems also send transactional history to the DMS through a scoped, idempotent API for centralized auditing. Staff use a web workspace to store, share, and audit internal documents. Every file is validated, scanned, and checksummed. The design needs no managed cloud services, keeping running costs low. Students are not DMS users.

3. Business context, objectives and scope

3.1 Organisation profile: The client is Pamantasan ng Lungsod ng Pasig, Pasig City, Philippines.

3.2 Business drivers:

- Four ERP systems keep separate databases; critical cross-system documents (like admission requirements) are not held in one controlled place.
- Evidence must be preserved: earlier file versions, received history, and audit entries cannot be overwritten.
- Integration must tolerate retries and out-of-order updates without duplicating or corrupting records.
- Running cost must stay low, so the design avoids managed database and storage services.

3.3 Objectives and success measures

ID	Objective	Measure	Target
OBJ-01	Centralized ERP document and history repository	Percentage of integrated ERP systems successfully sending documents and transaction history to the DMS	100%
OBJ-02	Safe retries and duplicate prevention	Number of duplicate history records created when an event is retried or resent	0
OBJ-03	Secure and trusted file storage	Number of files served without passing malware scanning or checksum verification	0
OBJ-04	Low operational and maintenance cost	Number of additional database engines required beyond the existing MySQL database	0
OBJ-05	Complete audit trail and accountability	Percentage of sensitive actions (uploads, downloads, modifications, deletions, and access changes) recorded in the audit log	100%

3.4 Scope

In scope	Out of scope
Document workspace with nested folders, version history, trash, restoration, and safe file previews	File preview support for formats outside of the agreed list
Document sharing with named recipients and comments, restricted to authorized users within the organization	External sharing, anonymous or public sharing links
Invitation-only account administration, approval of sensitive changes, and audit trail	Business approval or verification of admissions, enrollment, employment or payroll
/api/v1 boundary for history, documents, files and events	Source-side adapters and outbox in each ERP system

Integration with EMS, PMS, ES, and OAS for receiving transaction history and events	Permanent archival or custody of PMS (payroll), EMS (HR), and ES (enrollment) transactional documents
Storage of received transaction history and the latest snapshot of each source record	Direct management or modification of source records within ERP systems
DMS MySQL database and private version-file storage	Folder-level sharing and in-browser editing of document contents
Document custody and secure exchange of verified applicant documents from OAS to ES	Modification of business logic and databases within the ERP systems
Database-driven workspace registry for ERP and internal workspaces	Students and applicants: not DMS users, no DMS accounts

3.5 Justification for Centralized Document Custody

The Document Management System (DMS) serves as the centralized repository for institutional documents that require secure preservation, controlled access, version tracking, and authorized exchange between the school's ERP systems.

Although the ERPs maintain their own databases, their responsibilities primarily involve processing business transactions. The DMS provides a dedicated service for preserving the actual documents associated with those transactions, specifically when those documents must cross system boundaries.

The DMS is necessary for the following purposes:

1. Cross-system document exchange: Allow authorized ERP systems (specifically OAS and ES) to retrieve specific verified documents without requiring users to resubmit or manually transfer files.

2. Document integrity: Preserve immutable versions and SHA-256 checksums to verify that official documents have not been altered.

3. Controlled confidentiality: Restrict document access according to classification, originating workspace, and explicitly authorized recipients.

4. Auditability: Record document uploads, retrievals, modifications, releases, and access changes.

5. Reduced document duplication: Allow ERP systems to maintain DMS document identifiers and references instead of separately storing permanent copies of the same files.

Confidential documents shall only be transferred when there is a documented operational, archival, or cross-system requirement. The DMS shall not automatically ingest all records or attachments maintained by an ERP system.

3.6 ERP Document Custody and Business Justification

The DMS does not act as a duplicate permanent archive for every file generated by the ERP systems. Payroll (PMS), Human Resources (EMS), and Enrollment (ES) transactions remain fully within their respective databases. The DMS only holds custody of documents when there is a documented cross-system requirement or an independent institutional preservation need.

The OAS manages applicant submissions and verification. Documents submitted during admission are required by the ES when an admitted applicant proceeds to enrollment.

The DMS preserves the verified document versions and enables authorized retrieval by ES without requiring the applicant to submit the same evidence again. This eliminates manual file transfers and ensures the student's data is securely passed between systems.

Document exchange: OAS → DMS → ES

Only documents explicitly included in an authorized applicant packet shall be released to ES. Medical records are excluded from the ordinary OAS-to-ES exchange.

Proposed document inventory:

Document Type	**Purpose**	**DMS Justification**	**Classification**
PSA Birth Certificate	Identity and birth information verification	Preserve verified identity evidence for authorized enrollment retrieval	Confidential

Form 137 / Form 138	Academic record verification	Reuse verified academic evidence during enrollment	Confidential
Applicant Identification Document	Applicant identity verification	Preserve submitted identity evidence and revisions	Confidential
Parent/Guardian Identification Document	Parent or guardian verification	Retain required supporting evidence	Confidential
Guardianship Affidavit	Verification of legal guardianship	Preserve applicable guardianship evidence	Confidential
Applicant Photograph	Applicant identification	Preserve the submitted photograph associated with the application	Confidential

Signed Examination Ticket	Evidence of examination processing	Preserve the signed examination document where applicable	Confidential

3.6.2 Internal Office Workspaces (Non-ERP Documents)

While the four ERPs handle transactional records, school offices (e.g., Office of the Registrar, Dean's Office, Administrative Office) generate general institutional files that do not belong to a specific ERP database.

The DMS provides secure, access-controlled workspaces for these files, ensuring version history and audit trails are maintained without cluttering the transactional systems.

Document Type	Purpose	DMS Justification	Classification
Institutional Memorandums	Internal policy circulation	Centralized dissemination and audit of released versions	Internal

Meeting Minutes / Agendas	Record of official meetings	Immutable record of decisions and attendance	Internal
Policy Drafts and Approvals	Document versioning for institutional policies	Track revisions and final approvals	Internal
General Office Correspondence	Letters, requests, and external communications	Maintain an auditable trail of communications	Internal

4. Stakeholders and user personas

User counts are planning estimates, to be confirmed by the client. Peak load follows the system baseline of 100 concurrent browser sessions and 20 concurrent API writers. Applicants and students use their own ERP systems and never receive DMS accounts.

Persona	Approx. users (estimate)	Key tasks	Primary channel	Peak load
Super Admin (DMS system administrator)	2	Invite Admins; manage the workspace registry and connectors; oversee and alter DMS-held documents within the school and campus; approve Admin sensitive changes; review audit	Web	Within the 100-session baseline
Admin (office or college repository administrator)	5	Manage content and Staff within an assigned workspace; propose Staff invitations; approve Staff sensitive changes	Web	Within the 100-session baseline

Staff (faculty, college secretaries, department heads, deans and other school staff)	300	Keep own files; read shared memos; collaborate on assigned documents; request sensitive changes	Web	Within the 100-session baseline
ERP Representative (Staff in Registrar/Admission, HR and Payroll, affiliated to ES, OAS, EMS or PMS)	20	Store and review documents in their ERP workspace; open documents shared with them	Web	Within the 100-session baseline
ERP System Connector	4 systems	Send history and documents; read back history and events	API	20 concurrent writers; 120 requests per minute per connector

5. Requirements

Priorities follow MoSCoW (Must, Should, Could, Won't) and are proposed for client confirmation. A full list is held in the Business Requirements Document.

5.1 Functional requirements

ID	Module	Requirement	Priority
FR-01	Documents	Upload files as immutable versions after content validation and malware scan; keep every earlier version	Must
FR-02	Workspace	Nested folders, rename, move, star, trash and restore, safe previews, grid and list views	Must
FR-03	Sharing	Document sharing grants Viewer, Commenter or Editor permissions, with expiry of at most 30 days and revocation. Editors can rename the document and upload new versions (revision-checked). Editors cannot move, delete or reshare.	Must

FR-04	Comments	Comments on shared documents (up to 2,000 characters) and an activity sidebar	Should
FR-05	Accounts	Invitation-only accounts: Super Admin invites Admins; Admin proposes Staff within its workspace and Super Admin approves; the recipient sets their own password; Staff cannot invite; only Super Admin grants Admin or Super Admin privileges	Must
FR-06	History intake	Accept history events from EMS, PMS, ES and OAS; store an immutable receipt; update the latest snapshot only for newer revisions	Must
FR-07	History read-back	Cursor-based reads of history and events through the API; office and admin UI to inspect received versus current state	Must

FR-08	Audit	Record actor, action and document for sensitive actions	Must
FR-09	Scoped access	Limit users and connectors to their school, campus, workspace and, for Admin and Staff, their assigned memberships	Must
FR-10	Workspace registry	Database-driven registry of ERP and internal workspaces, each with a stable code, editable name and lifecycle state; a permission-filtered selector; adding or renaming a workspace needs no code change	Must
FR-11	Applicant packet	OAS releases specific verified document versions of an admitted applicant; ES retrieves only those Clean versions through an explicit grant; no medical files through the ordinary connector.	Must

FR-12	Approvals	Sensitive changes (sharing, classification, trash or restore, account changes): Staff requests go to a scoped Admin and Admin requests to Super Admin; approve or reject with a reason; no self-approval.	Must
FR-13	Document Type Registry	Maintain a database-driven registry of document types, originating workspaces, classifications, permitted formats, and authorized custody policies.	Must
FR-14	ERP Document Intake	Validate incoming documents against their registered type, authenticated originating workspace, permitted format, and required metadata before accepting them.	Must
FR-15	Controlled Document Exchange	Permit cross-ERP document retrieval only through an explicit, scoped, auditable release grant referencing authorized document versions.	Must

FR-16	Document Custody	Preserve immutable official documents under DMS custody while retaining business-record ownership in the originating ERP.	Must
ID	Module	Requirement	Priority

5.2 Non-functional requirements

ID	Category	Requirement
NFR-01	Security	Session sign-in with active-account check, CSRF protection and sign-in throttling; connector tokens stored hashed and bound to source, school and campus; HTTPS with TLS 1.2 or later and secure cookies; least privilege
NFR-02	Integrity	SHA-256 checksum recorded for every version and re-verified on access; versions are never changed
NFR-03	Idempotency	A retried request with the same key and content returns the original receipt (200). A changed request or stale revision returns 409. An older revision is stored (applied=false) without changing the latest snapshot.

NFR-04	Limits	API throttle 120 requests per minute per connector; file up to 10 MB; history payload up to 256 KiB; folder upload up to 50 files per batch; capacity baseline of 100 concurrent browser sessions and 20 concurrent API writers, to be validated by a load test on the target host
NFR-05	Availability	99.5% monthly service availability, excluding announced maintenance, measured by authenticated health checks
NFR-06	Recovery	RPO 24 hours and RTO 8 hours for single-site recovery; daily encrypted backup and quarterly restore drill; MySQL and private files restored together
NFR-07	Privacy	Compliant with the Philippine Data Privacy Act of 2012 (RA 10173); data residency to be agreed with the client and the Enterprise Architect
NFR-08	Auditability	Audit history kept for three years, online or archived; a legal hold suspends disposal
NFR-09	Accessibility	WCAG 2.2 Level AA target for staff-facing pages, with keyboard navigation and a skip link

6. Current state and gap analysis

Current system	Function	Data held	Pain point / gap	Disposition
Employee Management System (EMS)	Employee management	Own database (outside DMS)	Documents and change history not held in one controlled place; source adapter not yet built	Retain and integrate
Payroll Management System (PMS)	Payroll management	Own database (outside DMS)	As above	Retain and integrate
Enrollment System (ES)	Enrollment	Own database (outside DMS)	As above	Retain and integrate
Online Admission System (OAS)	Online admission	Own database (outside DMS)	As above	Retain and integrate

Source-side adapters are planned work for each ERP team. Data-quality checks, source identity matching and per-source payload schemas have not been done yet.

7. Target solution architecture

7.1 Architecture principles

- Clear Data Ownership: ERP systems manage their respective business records. The DMS manages cross-system documents, versions, sharing, audit logs, and received history.
- Unified Application Architecture: The DMS uses a single Laravel application organized into four logical tiers, without requiring separate servers or microservices.
- Controlled System Integration: All ERP systems communicate with the DMS through the /api/v1 endpoints, without direct access to each other's databases.
- Secure File Storage: Files are stored in private directories outside the publicly accessible web folder, with checksums used to verify file integrity.
- Reliable Data Processing: The system prevents duplicate records and handles simultaneous requests through request receipts, revision validation, and database constraints.
- Cost-Effective Infrastructure: The DMS uses a single MySQL database and a straightforward folder-based storage structure to minimize operational costs.

7.2 Logical architecture

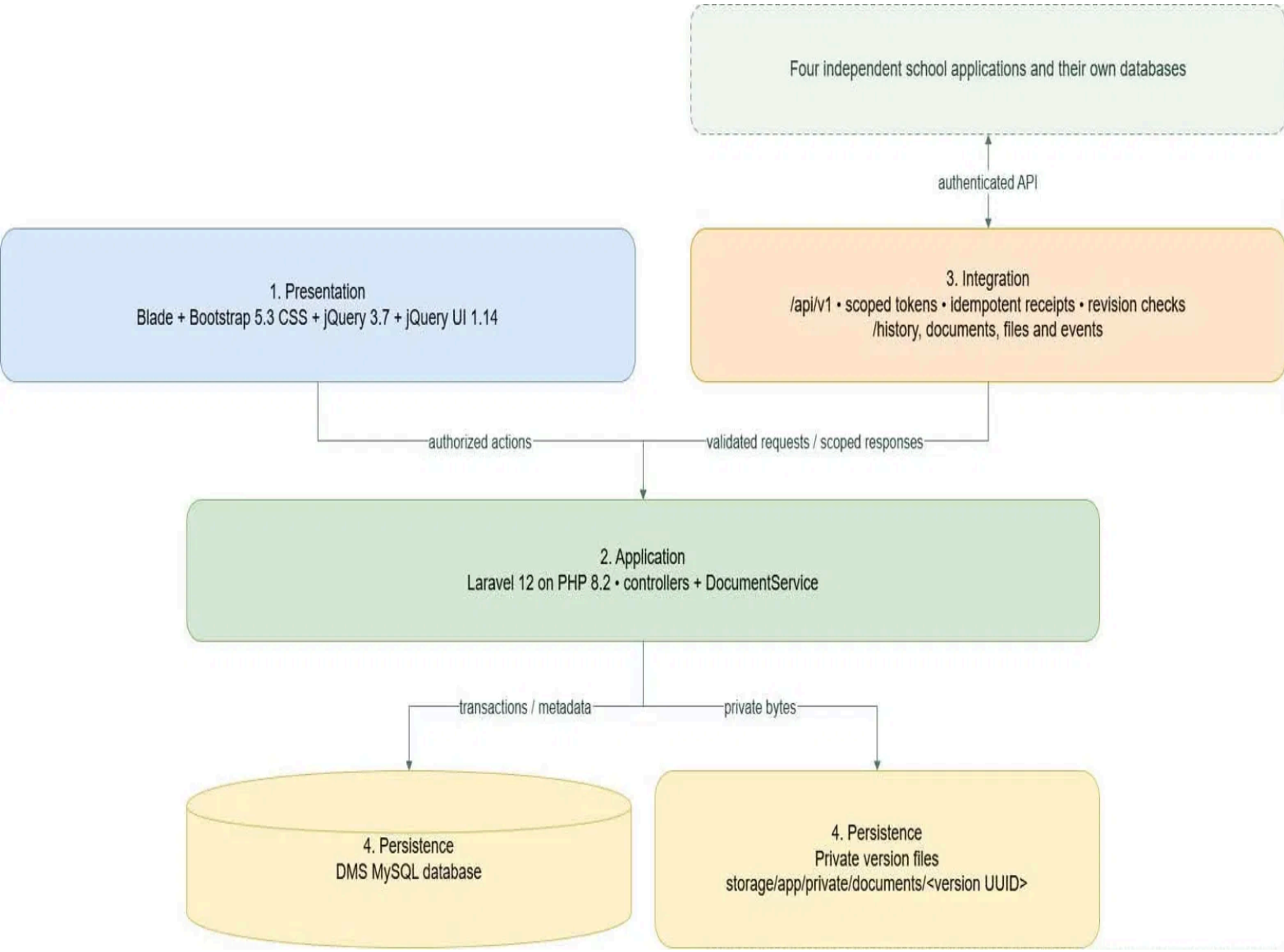


Figure 1: Logical architecture. Four tiers, with the four ERP systems outside DMS connecting only through tier 3.

Browser users reach the application through tier 1. The four ERP systems connect only through tier 3. Tiers 1 and 3 both feed tier 2, which is the only writer to the database and the private files.

7.3 Component responsibilities

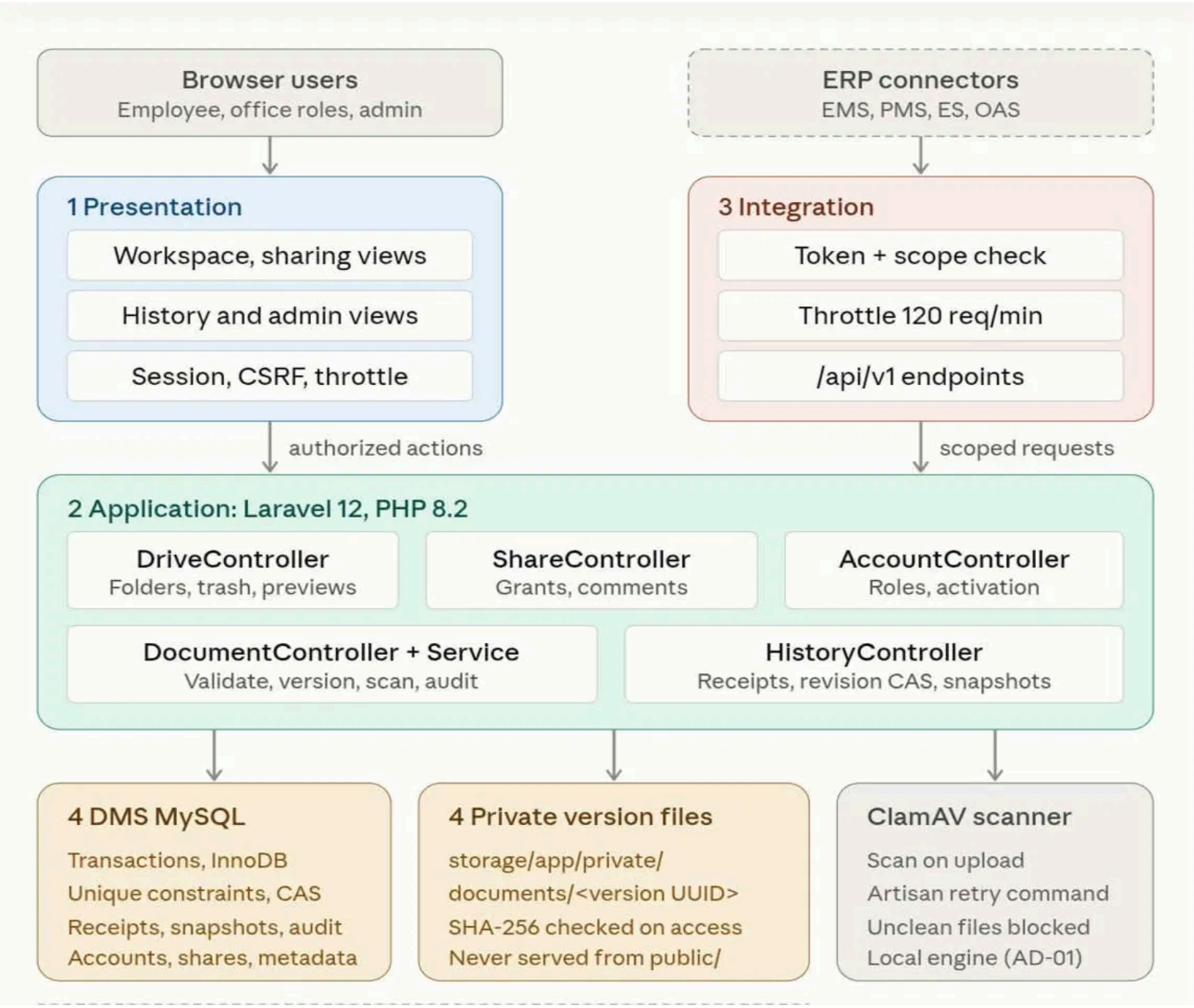


Figure 2: Component view. Controllers, throttle, ClamAV scanner and storage paths.

Component	Responsibility	Traces to
Presentation (Blade views, workspace)	Sign-in, role-aware navigation, folders, sharing and history screens	FR-02, FR-04, FR-07
Session security	Active account, CSRF, sign-in throttling, school/campus boundaries, grants	NFR-01, FR-09
DocumentController + DocumentService	Content validation, immutable versions, scan, idempotency, checksums, audit	FR-01, FR-08, NFR-02, NFR-03
DriveController	Folders, rename, move, star, trash and restore, previews, share creation and revocation	FR-02, FR-03
ShareController	Viewer, Commenter and Editor grants, pinned downloads, comments	FR-03, FR-04
AccountController	Invitation-only account creation, roles and activation, with approval by the next role up	FR-05

Workspace registry	Registered ERP and internal workspaces, lifecycle, memberships; feeds selectors and assignments	FR-09, FR-10
Approval queue	Pending requests, reviewer routing, authority re-check, atomic apply	FR-13
HistoryController	Canonical receipt hashing, revision ordering, record locks, snapshots	FR-06, FR-07, NFR-03
IntegrationAuth + API routes	Hashed token lookup, source/school/campus binding, throttle, /api/v1 endpoints	INT-01 to INT-05, INT-07, NFR-01, NFR-04
DMS MySQL database	Transactions, unique constraints, revision compare-and-swap	FR-06, NFR-03
Private storage + scanner	UUID paths, SHA-256, ClamAV scan and retry command	FR-01, NFR-02, INT-06

7.4 Build vs buy: All components are custom-built on Laravel 12. Malware scanning uses a local ClamAV engine instead of a paid scanning service (AD-01).

8. Data architecture and data protection

8.1 Core entities and system of record

Entity	Key attributes	System of record	Consumers
User account	id, email, password hash, role, active, school_id, campus	DMS	Workspace
Invitation	recipient email, role, workspace, token hash, status, inviter, approver	DMS	Admins, recipient
Workspace and membership	UUID, school/campus, code, name, kind (ERP or internal), lifecycle state; user, workspace, responsibilities	DMS	Selectors, assignments, access checks
Folder	UUID, parent_id, owner, name, source, school, campus, revision	DMS	Workspace

Document	UUID, owner, folder, title, subject, reference, source, category, classification, revision, trashed_at	DMS	Workspace; ERP systems (read)
Document version	UUID, document_id, number, filename, MIME type, size, checksum, storage_key, scan status	DMS (bytes in private storage)	Workspace, shares, ERP systems
Share, comment, star	document, version, recipient, permission, expiry, revocation	DMS	Recipients
Release grant	document or exact version, permitted user, group or connector, operation, expiry, revocation, release reference	DMS	ES (released packet), recipients
Approval request	requester, target, proposed change, reason, expected revision, reviewer, status	DMS	Approvers
Source record snapshot	scoped record key, source, school/campus, record type and ID, revision, deleted, payload	The source ERP system (DMS keeps the latest copy)	Office/admin history UI, API read-back

Received history receipt	cursor ID, record key, event ID, revision, request hash, operation, payload, applied	DMS (immutable)	Office/admin history UI, API read-back
Audit entry, integration event	school/campus, actor, action, document, event type, revision	DMS	Admins, ERP systems (events)
Connector identity, request receipt	source, school/campus, token hash; request key, request hash, response receipt	DMS	API

8.2 Data classification (proposed; each document carries a classification field and the classes need client sign-off)

Class	Examples	Controls
Restricted	Password hashes, token hashes, payroll and HR documents	Source-scoped access, private storage, audit
Confidential	Enrollment and admission documents, received history payloads, audit entries	School/campus and source scoping, private storage, checksum
Internal	General office working files	Owner and explicit grants

8.3 Retention and privacy

- Audit history is kept for three years, online or archived, and a legal hold suspends disposal. Retention for documents and received history is agreed with the client, aligned to the applicable records rules.
- A Data Protection Impact Assessment is completed before go-live.
- Data residency: production data and backups are stored in a region decided with the hosting model.
- ES and OAS payloads may contain personal data of students and applicants, who are not DMS users. Treat it as children's personal data and limit history access to office and admin roles.
- Payloads are source evidence, not account-provisioning instructions.
- Subject access and deletion requests: the process is not part of the current design and will be defined separately.

8.4 ERP Document Type Registry

The DMS shall maintain a database-driven document type registry identifying the documents authorized for intake from each ERP workspace.

Each registered document type shall contain:

Attribute	Description
Document Type Code	Unique identifier within the applicable registry scope
Document Type Name	Human-readable document name
Originating Workspace	ERP or internal workspace responsible for the document

Default Classification	Internal, Confidential, or Restricted
Permitted File Formats	Allowed file formats for the document type
Maximum File Size	File size restriction subject to the DMS system limit
Authorized Custodian	Office or role responsible for document custody
Authorized Recipients	Users, groups, or connectors eligible for an explicit access grant
Cross-System Release	Whether the document type is eligible for an approved ERP exchange
Retention Policy Reference	Applicable institutional records schedule
Lifecycle Status	Active, disabled for new intake, or retired

The document type registry shall be independent of the file extension and document classification.

A document type identifies the nature of the institutional record, while its file format identifies the technical representation of its contents.

8.5 Supported File Formats and Validation

The following file formats are proposed for the initial document ingestion policy.

Format	Extension	Intended Use	Preview
Portable Document Format	.pdf	Official documents, certificates, payslips, signed forms	Supported
JPEG Image	.jpg, .jpeg	Scanned identification documents and photographs	Supported
PNG Image	.png	Scanned forms, supporting images	Supported
Microsoft Word Document	.docx	Authorized office working documents	Download only
Microsoft Excel Workbook	.xlsx	Authorized office reports	Download only

All files shall be subject to the existing DMS controls:

- Maximum file size of 10 MB.
- Actual-content and MIME-type validation.
- Malware scanning through ClamAV.
- Private filesystem storage.
- SHA-256 checksum generation and verification.

- Immutable document versions.
- Access control enforcement before preview or download.

The DMS shall reject files that do not match the permitted formats of their registered document type.

8.6 Confidential Document Intake Policy

The DMS shall not automatically receive every confidential document stored in a connected ERP system.

A document type shall be eligible for DMS custody when at least one of the following conditions is satisfied:

1. The document requires independent institutional preservation.
2. Another authorized ERP system or institutional office requires controlled retrieval.
3. The document requires immutable version tracking and an independent audit history.
4. The institution has designated DMS as the authoritative custodian of the issued document.

Before enabling intake, the document type shall have a defined originating system, authorized custodian, operational purpose, classification, permitted recipients, and applicable retention policy.

Classification alone shall not authorize access. All retrievals shall be evaluated against the user's or connector's effective permissions and the document's explicit access grants.

Medical documents and other specially protected records shall not be accessible through ordinary ERP connector privileges or broad repository membership.

9. Integration architecture

All integrations go through the one /api/v1 boundary. No education data standard is used today; the contract is a JSON format carrying event ID, record identity, operation and revision.

ID	Interface	Direction	Pattern	Frequency	Data	Standard
INT-01	EMS	EMS → DMS	REST push, idempotent	Per record change	History events, documents	Bearer token, JSON
INT-02	PMS	PMS → DMS	REST push, idempotent	Per record change	History events, documents	Bearer token, JSON
INT-03	ES	ES → DMS	REST push, idempotent	Per record change	History events, documents	Bearer token, JSON
INT-04	OAS	OAS → DMS	REST push, idempotent	Per record change	History events, documents, applicant packet release	Bearer token, JSON

INT-05	ERP read-back	DMS → ERP (GET)	REST, cursor-based	On demand	History, events, documents, files	Bearer token, JSON
INT-06	Malware scanner	DMS → ClamAV	Command	Per upload and retry	File bytes	ClamAV command, Artisan retry
INT-07	Applicant packet read	DMS → ES (GET)	REST, read-only, explicit release grant	On demand after release	Allowed Clean document IDs and exact version IDs of a released packet	Bearer token, JSON

Error handling, retries and idempotency:

- New event: 201 with history_id, applied and duplicate. Identical replay: 200 with the original receipt. Same event with a different hash, or another event at the same record revision: 409.
- An older revision is stored with applied=false and the snapshot stays unchanged; a delete becomes a tombstone and earlier payloads are kept.
- Under contention DMS retries a bounded number of times, then returns 409. The source retries the same event with backoff and jitter.
- Throttle: 120 requests per minute per connector. Persistent conflicts need operator investigation; alert recipients are set in the support model.

9.1 ERP Document Exchange Policy

ERP systems shall exchange documents through authenticated /api/v1 interfaces. The DMS shall enforce the following:

1. Each document must identify its originating ERP workspace and registered document type.
2. The originating ERP must supply a stable external record reference.
3. DMS shall preserve the document and its immutable versions.
4. Cross-system access requires an explicit release grant.
5. A release grant shall identify the authorized recipient, permitted operation, and applicable document or exact version.
6. Every retrieval shall revalidate authorization, scan status, and file integrity.
7. All cross-system retrievals shall be recorded in the audit trail.
8. The receiving ERP shall not obtain general access to the originating ERP's document repository.
9. Revocation shall prevent subsequent retrievals without deleting historical evidence.

The initial and primary cross-system document exchange shall be the verified admission documents released from OAS to ES.

10. Security, identity and compliance

10.1 Identity and access

- Staff sign in with school accounts (email and password hash) in a session; inactive accounts are refused.
- Accounts are invitation-only: an individual, expiring, single-use link lets the recipient set their own password. Sensitive changes (sharing, classification, trash or restore, official publication, account changes) by Admin need Super Admin approval and by Staff need Admin approval; no self-approval.

- ERP systems authenticate with a bearer token stored hashed and bound to one source, school and campus.
- Single sign-on and multi-factor authentication are not in the current design; the Enterprise Architect decides whether to add them.

10.2 Role matrix

Role	Own files	Shared files	Workspace documents	Received history	Accounts and audit
Super Admin	Read, write (personal workspace)	Per grant	All DMS repositories within school and campus; access audited	Read	Manage; full scoped audit
Admin	Read, write	Per grant	Assigned workspace only; sensitive changes need Super Admin approval	Assigned source	Propose Staff invitations; assigned-source audit
Staff (including ERP Representatives)	Read, write	Per grant	Owned, assigned or explicitly shared documents; sensitive changes	None	Own request history only

			need Admin approval		
ERP connector	n/a	n/a	Own workspace, school, campus (API)	Write and read own	None

10.3 Security controls

- Transport: only the public/ directory is served, over HTTPS.
- Files: actual-content validation, 10 MB limit, private storage outside the web directory, UUID paths, SHA-256 checked on access, malware scan, and files that are not clean stay blocked.
- Application: CSRF protection, sign-in throttling, API throttle, hashed tokens, school/campus checks on every shared operation.
- Integrity: immutable versions and received history, canonical request hashes, unique constraints.
- Audit: audit entries with actor, action and document.
- Backups are encrypted (daily). Not yet decided: encryption at rest for live data, central monitoring, penetration testing.

10.4 Compliance checklist

- Data protection law: Philippine Data Privacy Act of 2012 (RA 10173)
- Personal data of minors inside ES and OAS payloads assessed in the DPIA
- TLS certificates and HTTPS-only access in place
- Backup and restore of MySQL plus private files tested
- Records retention rules for the client agreed
- Accessibility: WCAG 2.2 Level AA target for staff-facing pages

11. Infrastructure, deployment and operations

Status: in progress with the Enterprise Architect; not final. The hosting model is open. This section records what DMS needs whichever way it is hosted.

11.1 Hosting and environments

Environment	Purpose	Data	Hosting
Dev	Build and unit test	Synthetic	Pending Enterprise Architect decision
Test / SIT	Integration test with ERP adapters	Synthetic	Pending Enterprise Architect decision
UAT	Acceptance by reviewers and office users	Anonymised copy	Pending Enterprise Architect decision
Production	Live service	Real	Pending Enterprise Architect decision
Backup / DR	Recovery	MySQL dump and private files together	Pending Enterprise Architect decision

11.2 Platform (as designed)

- Compute: one PHP 8.2 / Laravel 12 application; only public/ is served.

- Data: one MySQL (InnoDB) database and a private folder for version files. No managed cloud database, queue, search or object storage is required, which keeps running cost low.
- Tenancy: school and campus scoping inside one deployment.
- Scanning: ClamAV command plus a scan-retry Artisan command.
- Delivery: CI/CD and environments are defined with the hosting model.

11.3 Operations and support

- Backups: MySQL and private files are backed up and restored together, because the database holds each file's storage_key and checksum. A daily encrypted backup is taken and a restore drill is run quarterly.
- Monitoring: authenticated health checks measure the 99.5% availability target; watch files stuck in Pending, Failed or Quarantined.
- Support tiers and SLAs: defined with the hosting model.
- The four ERP systems must be able to reach /api/v1; outages longer than their retry window lose delivery until they resend.
- Capacity baseline: 100 concurrent browser sessions and 20 concurrent API writers; a load test on the target host is still to be done.

12. Migration, implementation roadmap and testing

12.1 Delivery status and roadmap. Dates and phases are pending the Enterprise Architect.

Item	Status	Owner (proposed)
DMS workspace, document custody, sharing, audit (current baseline)	Implemented	DMS team

DMS /api/v1 boundary (history, documents, files, events)	Implemented	DMS team
Source adapter and outbox for EMS	Planned	EMS team
Source adapter and outbox for PMS	Planned	PMS team
Source adapter and outbox for ES	Planned	ES team
Source adapter and outbox for OAS	Planned	OAS team
Workspace registry and dynamic selector	Planned	DMS team
Invitation-only accounts and sensitive-change approvals	Planned	DMS team
OAS-to-ES verified applicant packet release	Planned	OAS, ES and DMS teams
Payload schemas and source identity matching	Not started	ERP teams with the DMS team

Full import (initial snapshots)	Not started	ERP teams (sending) and DMS team (receiving)
Retention schedule	Not started	Client
Tested restore	Not started	DMS team
Single sign-on	Not started	Enterprise Architect

12.2 Data migration approach

1. Agree record types and payload schemas with each ERP team.
2. Each source commits its local change and an outbox entry together, then sends a consistent initial snapshot with increasing revisions and a stable event ID.
3. DMS stores each snapshot and receipt; the source reconciles using cursor-based reads of /api/v1/history.
4. Run rehearsal loads and reconcile record counts per source with each reviewer.
5. Cut over per source in a window agreed with the Enterprise Architect.

12.3 Testing strategy

Test type	Scope	Owner	Exit criteria
Unit and integration	API contract: replay returns 200, altered hash returns 409,	DMS team	All critical paths pass

	out-of-order revision stored with applied=false		
Concurrency	Multi-writer MySQL, lock and unique-constraint behaviour at 100 concurrent sessions and 20 concurrent API writers	DMS team	No lost or duplicated history under load
Security	Upload validation, scan blocking, token and role scoping	DMS Solutions Architect	No open critical or high findings
Backup and restore	MySQL and private files restored together	DMS team	Restored files match stored checksums
Acceptance	End-to-end scenarios per ERP system and office role	ERP Solutions Architects, office users	Sign-off by each reviewer

12.4 Training and change management: role-based guides for office staff, and an /api/v1 integration guide for the four ERP teams, written once the contract is final.

13. Risks, assumptions, decisions and appendices

13.1 Risks (ratings proposed)

ID	Risk	Likelihood	Impact	Mitigation	Owner
R-01	Hosting model not settled	Medium	High	Decide with the Enterprise Architect; keep the design hosting-neutral	Enterprise Architect
R-02	ERP adapters late, or revisions not increasing	Medium	High	Publish the API contract; DMS stores out-of-order events as applied=false; reconcile by cursor	ERP Solutions Architects
R-03	MySQL and files not backed up or restored together	Medium	High	Joint backup and a tested restore	DMS Solutions Architect
R-04	Multi-writer MySQL contention untested	Medium	Medium	Load test; bounded retries; 409 handling	DMS team

R-05	Student or applicant data inside ES/OAS payloads	Medium	Very high	DPIA; restrict history to office and admin roles	Client / DPO
R-06	Scanner unavailable blocks uploads	Medium	Medium	Scan-retry command; monitor pending files	DMS team
R-07	Applicant packet release state is stale or ambiguous	Medium	High	Release by authenticated OAS events with revisions; fail closed; recheck the grant on every read	OAS and ES Solutions Architects
R-08	Medical files in OAS packets not yet confirmed	Medium	High	Confirm with the OAS owner before intake; keep them off the ordinary OAS and ES connector until access rules are agreed	Enterprise Architect

13.2 Assumptions and constraints

- Each ERP team builds its source adapter and keeps revisions increasing per record.
- DMS users are school staff; students have no access.

- The client and the Enterprise Architect decide hosting, budget and data residency.
- Each ERP solution architect reviews the integration contract for their system.

13.3 Architecture decisions log

ID	Decision	Options considered	Rationale	Date
AD-01	Use a local ClamAV engine, run through a scan-retry Artisan command, for malware scanning instead of commercial managed scanning APIs	Third-party SaaS (VirusTotal, AWS GuardDuty); ClamAV daemon / CLI	Aligns with the low operational cost requirement (OBJ-04); works on-premise or hosted; avoids external vendor recurring costs	2026-10-13
AD-02	One Laravel deployment, custom-built, with four logical tiers	Microservices; separate API service	Small operating footprint; shared DocumentService	2026-10-13
AD-03	One MySQL database and private folders	Managed cloud database; object storage	Lowest running cost; one backup set	2026-10-13

AD-04	Scoped bearer tokens, idempotent receipts, revision compare-and-swap	Direct database access; shared accounts	No writes to other systems' databases; safe retries	2026-10-13
AD-05	File scan is a security control, not business approval	Business verification inside DMS	Approval belongs to school users and originating systems	2026-10-13
AD-06	Students are not DMS users	Student accounts and workspaces	Outside the agreed scope	2026-10-13
AD-07	Hosting model	Local or on-premises; cloud	Open: pending the Enterprise Architect	Open
AD-08	Workspaces come from a database registry, not a fixed list of four	Fixed four-source list in code	A new ERP or internal workspace appears without code changes	2026-10-18
AD-09	Cross-system access only through explicit release grants	Same-school membership as access; broad connector tokens	Least privilege: ES reads only released versions	2026-10-18

AD-10	Invitation-only accounts, approved by the next role up	Self-registration; administrator-set passwords	Staff-only environment; the recipient owns their password	2026-10-18

13.4 Appendices

A. Glossary

Term	Meaning
DMS	Document Management System
EMS / PMS / ES / OAS	Employee Management, Payroll Management, Enrollment and Online Admission systems
Source	One of the four ERP systems when it sends history to DMS
Workspace	A registered repository for an ERP system or an internal office
Release grant	An explicit permission to read a document or an exact version, given to a named recipient or connector
Receipt	Stored result of an API request, returned again on an identical retry

Revision	Per-record number from the source that only increases
CAS	Compare-and-swap: update only if the revision is newer
DPIA	Data Protection Impact Assessment
RPO / RTO	Recovery point and recovery time objectives

B. Requirements traceability (requirement to component to test)

Requirement	Component	Test
FR-01, NFR-02	DocumentService, private storage, scanner	Unit and integration; security
FR-02, FR-03, FR-04	DriveController, ShareController	Acceptance
FR-05, FR-09, NFR-01	AccountController, invitations, session security, IntegrationAuth	Security
FR-06, FR-07, NFR-03	HistoryController, MySQL	Unit and integration; concurrency

FR-08	DocumentService, audit entries	Acceptance
FR-10	Workspace registry, MySQL	Unit and integration; acceptance
FR-11	DocumentService, ShareController	Acceptance
FR-12	IntegrationAuth, HistoryController, release grants	Security; acceptance
FR-13	Approval queue, AccountController	Security; acceptance
NFR-06	Backup of MySQL and private files	Backup and restore